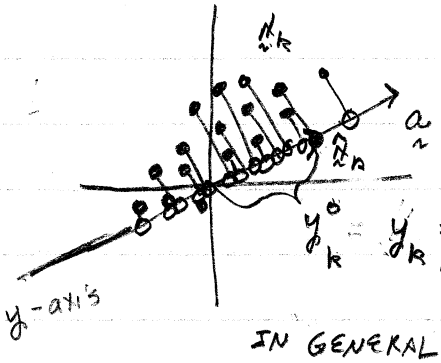
STATISTICS 652
LECTURE 8 - 2026

Visualization of a one dimensional projection of a multivariate data set.

Consider a q-dimensional data set $\{\underset{\sim}{x}_1, \ldots, \underset{\sim}{x}_n\}$ and the one dimensional data set $\{y_1 = \underset{\sim}{a}'\underset{\sim}{x}_1, \ldots, y_n = \underset{\sim}{a}'\underset{\sim}{x}_n\}$



$y_k^0 = y_k = \underset{\sim}{a}'\underset{\sim}{x}_k$ when $\|\underset{\sim}{a}\| = 1$

IN GENERAL, $y_k^0 = \underset{\sim}{b}'\underset{\sim}{x}_k$ where $\underset{\sim}{b} = \underset{\sim}{a}/\|\underset{\sim}{a}\|$

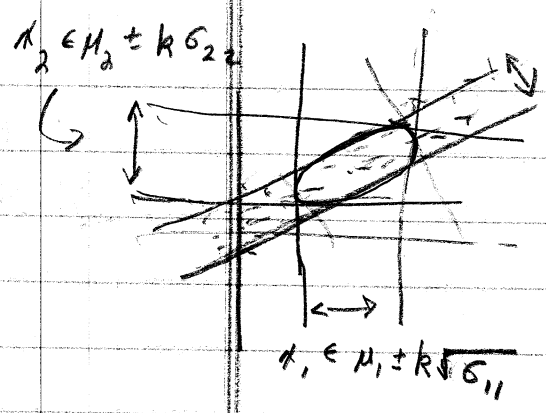
OR $y_k = y_k^0 \cdot \|\underset{\sim}{a}\|$

ORTHOGONAL PROJECTION of $\underset{\sim}{x}_k$ ONTO $\underset{\sim}{a}$ IS $\hat{\underset{\sim}{x}}_k = y_k^0 \underset{\sim}{b} = \dfrac{\underset{\sim}{a}\underset{\sim}{a}'}{\|\underset{\sim}{a}\|^2} \cdot \underset{\sim}{x}_k$

RECALL LEMMA 7.6. GIVEN $\underset{\sim}{x} \in \mathbb{R}^q$, define $z_{\underset{\sim}{a}} = \dfrac{\underset{\sim}{a}'\underset{\sim}{x} - \underset{\sim}{a}'\underset{\sim}{\mu}}{\sqrt{\underset{\sim}{a}'\Sigma\,\underset{\sim}{a}}}$, i.e. z-score for $y = \underset{\sim}{a}'\underset{\sim}{x}$. THEN $\sup_{\substack{\underset{\sim}{a} \in \mathbb{R}^q \\ \|\underset{\sim}{a}\| \neq 0}} |z_{\underset{\sim}{a}}| = D_q(x) \equiv \sqrt{(\underset{\sim}{x} - \underset{\sim}{\mu})'\Sigma^{-1}(\underset{\sim}{x} - \underset{\sim}{\mu})}$

with supremum obtained ONLY AT $\underset{\sim}{a} \propto \Sigma^{-1}(\underset{\sim}{x} - \underset{\sim}{\mu})$.

NOTE: $\sup_{\substack{\underset{\sim}{b} \in \mathbb{R}^q \\ \|\underset{\sim}{b}\| = 1}} |z_{\underset{\sim}{b}}| = D_q(x)$ SINCE $\|\underset{\sim}{a}\|$ CANCELS IN NUMERATOR AND DENOMINATOR WHEN SETTING $\underset{\sim}{b} = \underset{\sim}{a}/\|\underset{\sim}{a}\|$,



CONSIDER THE INTERSECTION OF STRIPS

$$\underset{\sim}{b}'\underset{\sim}{N} \in \underset{\sim}{b}'\underset{\sim}{\mu} \pm k\sqrt{\underset{\sim}{b}'\underset{\sim}{\Sigma}\underset{\sim}{b}}$$

OVER ALL $\underset{\sim}{b} \in \mathbb{R}^2$, $\|\underset{\sim}{b}\| = 1$

LEMMA 8.1. THE INTERSECTION OF STRIPS

$$a = \bigcap_{\substack{\underset{\sim}{a} \in \mathbb{R}^2 \\ \underset{\sim}{a} \neq 0}} \left\{ \underset{\sim}{N} \in \mathbb{R}^2 \;\middle|\; \underset{\sim}{a}'\underset{\sim}{N} \in \underset{\sim}{a}'\underset{\sim}{\mu} \pm k\sqrt{\underset{\sim}{a}'\underset{\sim}{\Sigma}\underset{\sim}{a}} \right\} = \bigcap_{\substack{\underset{\sim}{b} \in \mathbb{R}^2 \\ \|\underset{\sim}{b}\| = 1}} \left\{ \underset{\sim}{N} \in \mathbb{R}^2 \;\middle|\; \underset{\sim}{b}'\underset{\sim}{N} \in \underset{\sim}{b}'\underset{\sim}{\mu} \pm k\sqrt{\underset{\sim}{b}'\underset{\sim}{\Sigma}\underset{\sim}{b}} \right\}$$

$$= \left\{ \underset{\sim}{N} \in \mathbb{R}^2 \;\middle|\; (\underset{\sim}{N}-\underset{\sim}{\mu})'\underset{\sim}{\Sigma}^{-1}(\underset{\sim}{N}-\underset{\sim}{\mu}) \leq k^2 \right\}$$

$$= \left\{ \underset{\sim}{N} \in \mathbb{R}^2 \;\middle|\; D_2(\underset{\sim}{N}) \leq k \right\}$$

PROOF: FOLLOWS FROM LEMMA 7.6

NOTES: i) a REPRESENTS AN ELLIPSE AND ITS INTERIOR.

ii) ALL POINTS IN a LIE WITHIN k S.D. OF ITS MEAN IN ANY UNIVARIATE PROJECTION, i.e. HAVE ABSOLUTE z SCORES $\leq k$.

iii) ALL POINTS NOT IN a LIE MORE THAN k S.D. FROM ITS MEAN IN SOME PROJECTION.

DEFINITION 8.2. GIVEN $\Sigma > 0$, THE MAHALANOBIS INNER PRODUCT ON $\mathbb{R}^q$ IS DEFINED AS $\langle x, y \rangle_{\Sigma^{-1}} = x' \Sigma^{-1} y$. HENCE, THE MAHALONOBIS NORM IS $\|x\|_{\Sigma^{-1}} = \sqrt{x' \Sigma^{-1} x}$. ALSO THE MAHALONOBIS DISTANCE $D_q(x) = \|x - \mu\|_{\Sigma^{-1}}$.

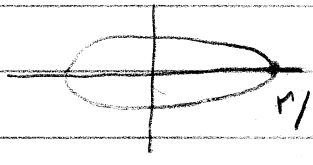
NOTE: THE ORBITS $\{x \mid D_q(x) = k\}$ ARE CONCENTRIC ELLIPSES ABOUT $\mu \in \mathbb{R}^q$.

REVIEW: EQUATION OF AN ELLIPSE IN $\mathbb{R}^q$

$x \in \mathbb{R}^q$ i) $x'x = r^2$ SPHERE ABOUT ORIGIN WITH RADIAS r

ii) $\sum_{j=1}^{q} \alpha_j x_j^2 = r^2$ ELLIPSE. AXIS: $e_1, \ldots, e_q$, i.e, EUCLIDEAN BASIS

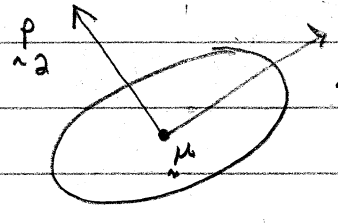
HALF LENGTHS: $\frac{r}{\alpha_1}, \ldots, \frac{r}{\alpha_q}$ $\alpha_j \geq 0$



$\Rightarrow x' \Delta x = r^2$, $\Delta = \text{diag}(\alpha_1, \ldots, \alpha_q)$

NOTE: $\alpha_j = 0 \Rightarrow$ ELLIPSE WITH INFINITE AXIS

iii) GENERAL ELLIPSE



$P = [\rho_1, \rho_2 \cdots \rho_q]$ ORTHONORMAL COLUMNS, P: ORTHOGONAL MATRIX

$y = P'(x - \mu)$

$r^2 = y' \Delta y = (x-\mu)' P \Delta P' (x - \mu) = (x-\mu)' A (x - \mu)$

where $A = P \Delta P' \geq 0$, and $\Delta = \text{diag}(\alpha_1, \ldots, \alpha_q)$

AXIS: $\rho_1, \ldots, \rho_q$.

HALF LENGTHS: $\frac{r}{\alpha_1}, \ldots, \frac{r}{\alpha_q}$

REVIEW OF SOME LINEAR ALGEBRA

ORTHOGONAL MATRICES:

A $q \times q$ MATRIX P IS ORTHOGONAL IFF $PP' = I_q$.

PROPERTIES: ORTHOGONAL MATRICES FORM A GROUP, SAY $\mathcal{O}(p)$, under MATRIX MULTIPLICATION. THAT IS

i) $I \in \mathcal{O}(p)$ ii) $P^{-1} = P' \in \mathcal{O}(p)$ IDENTITY + INVERSE

iii) $P_1, P_2 \in \mathcal{O}(p) \Rightarrow P_1 P_2 \in \mathcal{O}(p)$ and CLOSURE

iv) $P_1, P_2, P_3 \in \mathcal{O}(p) \Rightarrow (P_1 P_2) P_3 = P_1 (P_2 P_3)$. ASSOCIATIVE

HENCE, by (ii) BOTH THE ROWS AND COLUMNS OF P ARE ORTHONORMAL.

ALSO, $\det(P) = \pm 1$ SINCE $1 = \det(I) = \det(PP') = \det(P)\det(P')$
$= \{\det(P)\}^2$,

NOTE: AS A TRANSFORMATION, P IS EITHER a) ROTATION, b) REFLECTION, c) PERMUTATION OR A COMBINATION OF THESE.

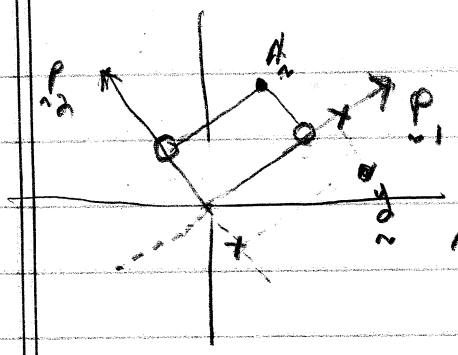
- P preserves lengths and angles, i.e. $\langle \underset{\sim}{x}, \underset{\sim}{y} \rangle = \langle P\underset{\sim}{x}, P\underset{\sim}{y} \rangle$.

THEOREM 8.3, THE SPECTRAL VALUE DECOMPOSITION,

For a $q \times q$ symmetric matrix A, there exists an orthogonal matrix $P = [\underset{\sim}{P}_1 \cdots \underset{\sim}{P}_q]$ and a diagonal matrix $\Delta = \mathrm{diag}(\lambda_1, \ldots, \lambda_p)$ with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_q$ such that

$$A = P\Delta P' = \sum_{j=1}^{q} \lambda_j \underset{\sim}{P}_j \underset{\sim}{P}_j' = \sum_{j=1}^{q} \lambda_j \underset{\sim}{P}_j \quad \text{where } \underset{\sim}{P}_j = \underset{\sim}{P}_j \underset{\sim}{P}_j'$$

NOTE: $\underset{\sim}{P}_j$ is the orthogonal projection operator onto the one dimensional subspace spanned by $\underset{\sim}{P}_j$.



As a linear transformation

$$y = Ax = P\Delta P' x = \sum_{j=1}^{q} \lambda_j p_j p_j' x$$

i.e. A ACTS AS a diagonal transformation ON THE AXES $p_1, \ldots, p_q$.

"UNIQUENESS" of S, V, D. For a symmetric matrix

$A = P\Delta P'$ IFF $Ap_j = \lambda_j p_j$ i.e. λ_j are the ordered eigenvalues of A and p_j are the associated eigenvectors (normalized)

SELF REVIEW: PROOF OF S, V, D (ALSO, SEE HW PROBLEMS)

PROPERTIES: FOR A SYMMETRIC MATRIX A,

i) $A \geq 0$ IFF $\lambda_1 \geq \cdots \geq \lambda_q \geq 0$ $(\Delta \geq 0)$

ii) $A > 0$ IFF $\lambda_1 \geq \cdots \geq \lambda_q > 0$. $(\Delta > 0)$

iii) FOR $A \geq 0$, $A = BB'$ where $B = P\Delta^{1/2}Q'$ with $Q \in O(q)$.
B is a square root of A.

GENERAL ELLIPSE:

$$(x - \mu)' A (x - \mu) = r^2 \quad \text{with } A = P\Delta P'$$

IS AN ELLIPSE IFF $A \geq 0$ $(\Delta \geq 0)$

a) CENTER: μ

b) AXIS FROM μ: $p_1, \ldots, p_q$

c) RADII: $\frac{r}{\sqrt{\lambda_1}}, \ldots, \frac{r}{\sqrt{\lambda_q}}$ $\qquad r = \sqrt{r^2} \geq 0$

NOTE: IF $A \not> 0$, IT IS A HYPERBOLA OR MIXTURE OF HYPERBOLA + ELLIPSE.

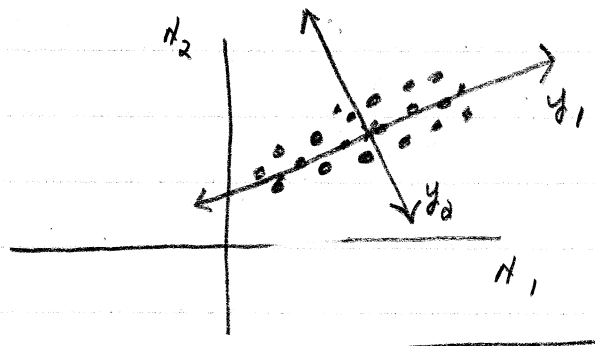
PRINCIPAL COMPONENTS TRANSFORMATION

$$\underset{\sim}{X} \in \mathbb{R}^q, \quad \underset{\sim}{\mu} = E(\underset{\sim}{X}), \quad \Sigma = \mathrm{VAR}(\underset{\sim}{X}) = P \Delta P' \geq 0$$

PRINCIPAL COMPONENTS: $\underset{\sim}{Y} = P'(\underset{\sim}{X} - \underset{\sim}{\mu})$

$$\Rightarrow E[\underset{\sim}{Y}] = \underset{\sim}{0} \quad \text{and} \quad \Sigma_Y = \mathrm{VAR}(Y) = P' \Sigma P = \Delta$$

THAT IS, THE P.C. VARIABLES $\underset{\sim}{Y}$ ARE A TRANSLATION FOLLOWED BY AN ORTHOGONAL TRANSFORMATION (HENCE RIGID, I.E. NON-DISTORTING) OF THE ORIGINAL VARIABLES $\underset{\sim}{X}$ TO UNCORRELATED MARGINALS WITH NON-INCREASING VARIANCES



NOTE: $D_q(\underset{\sim}{x}; \underset{\sim}{\mu}, \Sigma) = \sqrt{(\underset{\sim}{x} - \underset{\sim}{\mu})' \Sigma^{-1} (\underset{\sim}{x} - \underset{\sim}{\mu})} = k \quad \text{for } \Sigma > 0$

$$\Leftrightarrow D_q(\underset{\sim}{y}; \underset{\sim}{\mu}_Y, \Sigma_Y) = \sqrt{\underset{\sim}{y}' \Delta^{-1} \underset{\sim}{y}} = k$$

FOR PC:
- CENTER: 0
- AXIS: $\underset{\sim}{y}_1, \ldots, \underset{\sim}{y}_q$
- RADII: $k\sqrt{\lambda_1}, \ldots, k\sqrt{\lambda_q}$

IMPORTANT PROPERTY OF PCA:

PCA IS DEPENDENT ON SCALE

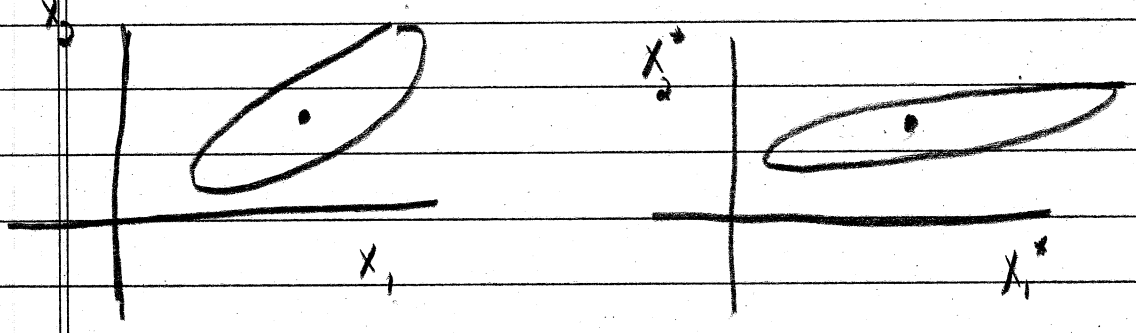
e.g. $X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$ FEET / OUNCES and $X^* = \begin{pmatrix} X_1^* \\ X_2^* \end{pmatrix}$ INCHES / POUNDS,

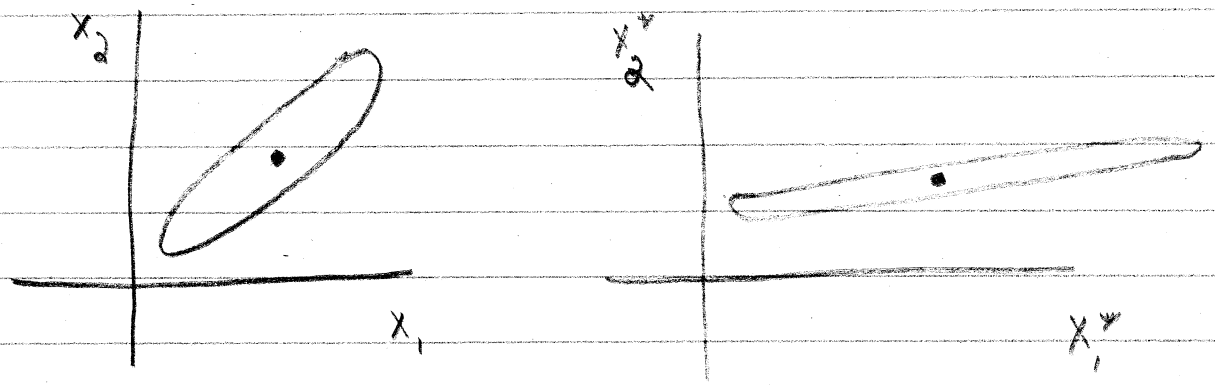
i.e. $\underset{\sim}{X}^* = \begin{pmatrix} 12 & 0 \\ 0 & 1/16 \end{pmatrix} \underset{\sim}{X} = D\underset{\sim}{X}$

THEN, THERE IS NO SIMPLE RELATIONSHIP BETWEEN

PCA FOR $\underset{\sim}{X}$, with $\Sigma_X = P\Delta P'$

AND FOR $\underset{\sim}{X}^*$, with $\Sigma_{X^*} = D\Sigma_X D' = P^*\Delta^* P^{*\prime}$.





IN PRACTICE:

- APPLY PCA WHEN MEASUREMENTS HAVE THE SAME UNITS

  e.g. • GROWTH CURVE (VARIABLES MEASURE AN INDIVIDUALS HEIGHT OVER TIME)

  • VALUE OF STOCKS

  • SPECTRUM OF CHEMICAL COMPOUND (VARIABLES CORRESPOND TO INTENSITY AT DIFFERENT FREQ.)

- APPLY PCA TO INDIVIDUALLY STANDARDIZED VARIABLES, i.e. TO THE CORRELATION MATRIX

$$R = D(\Sigma)^{-\frac{1}{2}} \Sigma D(\Sigma)^{-\frac{1}{2}} \quad \text{where } D(\Sigma) = \text{diag}(\sigma_1^2, \dots, \sigma_q^2)$$

$$= \{\rho_{ij}\} \qquad \rho_{ij} = \frac{\sigma_{ij}}{\sigma_i \sigma_j}$$

NOTE: FOR $q = 2$, $R = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \Rightarrow \lambda_1 = 1 \pm \rho$ with

$$R\underset{\sim}{a} = (1+\rho)\underset{\sim}{a} \Rightarrow \underset{\sim}{a} \propto \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ and}$$

$$R\underset{\sim}{a} = (1-\rho)\underset{\sim}{a} \Rightarrow \underset{\sim}{a} \propto \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

MINIMAX PROPERTIES OF EIGENVALUES/EIGENVECTORS FOR RAYLEIGH QUOTIENTS $\underset{\sim}{x}'A\underset{\sim}{x}/\underset{\sim}{x}'\underset{\sim}{x}$.

THEOREM 8.4. Let $A_{q\times q}$ be symmetric with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_q$, then

i) $\sup_{\substack{\underset{\sim}{x}\in\mathbb{R}^q \\ \underset{\sim}{x}\neq\underset{\sim}{0}}} \frac{\underset{\sim}{x}'A\underset{\sim}{x}}{\underset{\sim}{x}'\underset{\sim}{x}} = \lambda_1$ with sup obtained IFF $A\underset{\sim}{x} = \lambda_1\underset{\sim}{x}$.

ii) $\inf_{\substack{\underset{\sim}{x}\in\mathbb{R}^q \\ \underset{\sim}{x}\neq\underset{\sim}{0}}} \frac{\underset{\sim}{x}'A\underset{\sim}{x}}{\underset{\sim}{x}'\underset{\sim}{x}} = \lambda_q$ with inf obtained IFF $A\underset{\sim}{x} = \lambda_q\underset{\sim}{x}$.

PROOF: APPROACH 1: CALCULUS (USING LAGRANGE MULTIPLIER)

APPROACH 2: USING S.V.D. $A = P\Delta P'$

$$\frac{\underset{\sim}{x}'A\underset{\sim}{x}}{\underset{\sim}{x}'\underset{\sim}{x}} = \frac{\underset{\sim}{x}'P\Delta P'\underset{\sim}{x}}{\underset{\sim}{x}'\underset{\sim}{x}} = \frac{\underset{\sim}{y}'\Delta\underset{\sim}{y}}{\underset{\sim}{y}'\underset{\sim}{y}} \quad \text{where } \underset{\sim}{y} = P'\underset{\sim}{x}$$

$$= \frac{\sum_{j=1}^q \lambda_j y_j^2}{\sum_{j=1}^q y_j^2} = \sum_{j=1}^q \kappa_j \lambda_j \quad \text{where } \kappa_j = \frac{y_j^2}{\sum_{i=1}^q y_i^2}$$

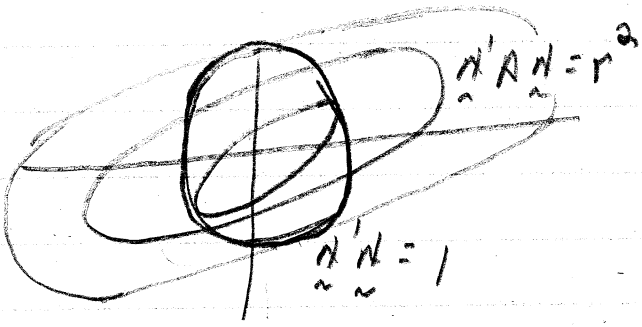
and so $\kappa_j \geq 0$, $\sum_{j=1}^q \kappa_j = 1$ (CONVEX COMBINATION OF $\lambda_1,\ldots,\lambda_q$)

$\leq \lambda_1$ with equality if $\kappa_1 = 1$, $\kappa_j = 0$ $j = 2,\ldots,q$

$\geq \lambda_q$ with equality if $\kappa_q = 1$, $\kappa_j = 0$ $j = 1,\ldots,q-1$

ALSO NOTE IF $A\underset{\sim}{x} = \lambda_j\underset{\sim}{x}$ then $\underset{\sim}{x}'A\underset{\sim}{x}/\underset{\sim}{x}'\underset{\sim}{x} = \lambda_j$. ∎

INTUITIVE DERIVATION : FOR $A > 0$



IF $A \not> 0$, REPLACE WITH

$$A_0 = A - (\lambda_2 + \varepsilon) I > 0$$

GENERALIZATION :

i) $\sup_{\underset{\sim}{x} \neq 0}$ $\quad \underset{\sim}{x} \perp \{\underset{\sim}{p}_1, \ldots, \underset{\sim}{p}_k\}$

$$\frac{\underset{\sim}{x}'A\underset{\sim}{x}}{\underset{\sim}{x}'\underset{\sim}{x}} = \lambda_{k+1}$$

with equality IFF

$$A\underset{\sim}{x} = \lambda_{k+1}\underset{\sim}{x}$$

ii) $\inf_{\underset{\sim}{x} \neq 0}$ $\quad \underset{\sim}{x} \perp \{\underset{\sim}{p}_{k+1}, \ldots, \underset{\sim}{p}_q\}$

$$\frac{\underset{\sim}{x}'A\underset{\sim}{x}}{\underset{\sim}{x}'\underset{\sim}{x}} = \lambda_k$$

with equality IFF

$$A\underset{\sim}{x} = \lambda_k \underset{\sim}{x}$$